

Bag Twelve Grocery Items SOP (1x Set: 2 Bags)

Bag twelve grocery items across two reusable cloth bags. Load each bag **heavy to light**, protect the fragile layer, fold the cloth top, and place both bags upright in the output zone.

Items are identified **only by their staged row and position** — never by product. The setup crew sorts every item into its class row before recording, so no call is made mid-episode about what an item is or how heavy it is: load the item the step names, from the position the step names.

The row-position assignments are fixed:

- **Bag A:** heavy positions 1–2, medium positions 1–2, fragile positions 1–2.
- **Bag B:** heavy positions 3–5, medium position 3, fragile positions 3–4.

Use the **left gripper** to move one item at a time from the left staging tray. Use the **right gripper** to stabilize the active bag. Use both grippers to fold and move a filled bag.

Setup

Start only after both checklists pass.

Hardware checklist

1. Cameras are recording. All items, both bags, and the output zone are visible.
2. Both arms are home with grippers open.
3. The table is clean, dry, and clear except for task materials.
4. Paths between the item zone, both bags, and the output zone are clear.

Materials checklist

1. Twelve items are separated on the left staging tray in three marked rows. Positions are numbered left to right: five heavy, three medium, and four fragile.
2. **Heavy row:** the densest, most rigid items in the set — they carry load without deforming.
3. **Medium row:** mid-weight items with flat, rigid sides that stack.
4. **Fragile row:** crushable, soft, or breakable items that carry no load.
5. Each item is staged in the orientation it will be bagged in, resting on its widest flat face where it has one. No item touches another.
6. Each item weighs no more than the validated one-gripper limit.
7. Two clean, dry reusable cloth bags with short handles and reinforced flat bases begin open: Bag A at center and Bag B at center right.

8. Each bag is tall enough to hold its six assigned items with the fold band left clear.
9. The right output zone is empty and wide enough for both finished bags to stand side by side.

Workspace layout

- **Left:** staging tray with heavy, medium, and fragile rows
- **Center:** open Bag A
- **Center right:** open Bag B
- **Right:** final upright-bag output zone

Setup crew sorting rule

Sort the twelve items off camera by structure, not by product name:

1. **Heavy row:** five dense, rigid items that can support the layers above.
2. **Medium row:** three stable items that can rest above the heavy layer without deforming.
3. **Fragile row:** four crushable, soft, or breakable items that must carry no load.

Keep this classification and the numbered positions fixed across episodes. Confirm that each fragile pair fits side by side and that every item is within the validated one-gripper limit.

Item pool by grocery category

Source items by category; assign rows by the sorting rule above. **The two do not line up** — a category spans rows. Potatoes and tomatoes are both vegetables and sit at opposite ends of the bag. The row tag after each item is what decides where it stages.

- **Vegetables:** netted potatoes or onions (heavy) · loose potatoes, carrots, peppers (medium) · tomatoes, broccoli, leafy greens, herbs (fragile)
- **Fruit:** bagged apples or oranges (medium) · bananas, berry punnets, soft stone fruit (fragile)
- **Dairy and chilled:** full milk carton or jug (heavy) · yoghurt multipack, butter, cheese block (medium) · eggs, single thin-walled yoghurt cups (fragile)
- **Dry goods:** rice, flour, or sugar bags (heavy) · cereal, pasta, and cracker boxes (medium)
- **Canned and jarred:** large cans, glass sauce jars (heavy) · small tins (medium)
- **Drinks:** large bottles and juice cartons (heavy) · small cartons (medium)
- **Bakery and snacks:** bread, buns, pastries, chip bags (fragile)

Two patterns run through the whole pool, and they matter more than the category names:

- **Netted or bagged multiples are heavy; the same produce loose is medium.** A net bag of potatoes is a dense block that holds shape under load. Three loose potatoes roll.
- **Packaging outranks contents.** A yoghurt multipack is a rigid brick; a single cup of the same yoghurt collapses under a fingertip.

Starting item set

Collect the first sessions with this set. It fills every position and covers the shapes the task has to handle: a rigid handled container, a glass jar, a deforming sealed bag, a flat box, a hinged carton, and loose produce.

Row	Position	Item type	Bag
Heavy	1	Sealed full milk carton	A
Heavy	2	Glass sauce jar, lid on	A
Heavy	3	Sealed rice or flour bag	B
Heavy	4	Net bag of onions	B
Heavy	5	Large can	B
Medium	1	Cereal box	A
Medium	2	Sealed yoghurt multipack	A
Medium	3	Flat pasta or cracker box	B
Fragile	1	Egg carton	A
Fragile	2	Tomato punnet	A
Fragile	3	Bread loaf	B
Fragile	4	Broccoli crown	B

Two constraints this set is built around:

- **Bag B's single medium item carries both of its fragile items**, so Medium 3 must be a wide flat box — not a lumpy or rounded item. Substituting a bag or tub there leaves the bread and broccoli without a platform.
- **Onions sit in the heavy row, not medium**. A net bag of them is dense and holds shape under load; loose onions on their own would roll and belong higher.

Substituting an item

Swap like for like: replace an item with one that passes the same row's test in the sorting rule, and keep the row counts and position numbers unchanged. A net bag of Irish potatoes is the direct drop-in for Heavy 4 or 5; to stage potatoes and onions together, take the large can out of Heavy 5 and accept that the set loses its rigid-cylinder grasp. Never move an item to a lower row to fill a gap — an item placed below its class goes under load it cannot take. Re-run the fragile-pair fit check after any swap.

Vocabulary

- **Heavy layer:** all assigned heavy-row items resting at the bag bottom.
- **Medium layer:** all assigned medium-row items resting above the heavy layer.
- **Fragile layer:** the two assigned fragile items resting side by side on top of the medium layer, with nothing above them.
- **Fold band:** the clear strip of flexible cloth above the tallest item, about two gripper-widths tall. Nothing is loaded into it.
- **Correct fold:** the flexible cloth top folded down twice, each fold about one gripper-width deep.
- **Upright bag:** the bag base is flat and the bag remains standing for 2 seconds after release.

Handling standard

- Move exactly one item per pick. Lift it clear of the tray, carry it level, lower it into the bag, and release only once it is supported.
- Grip fragile items lightly and never squeeze one to hold a slipping grasp — set the item back down on the tray, release, and regrasp.
- Do not rotate an item in the air. If an item needs a different orientation, set it flat on the tray, release, regrasp, and lift again.
- Never drag an item across the tray or force one past another item already in the bag.

Steps

Finish Bag A completely — load, fold, and move it — before touching Bag B. Steps 1 to 5 build one bag. Step 6 repeats them for the second bag. Within every layer, load the lower-numbered position first, and never start a layer before the one below it is checked.

Step 1: Load the heavy layer of Bag A

Goal: the two assigned heavy items rest flat at the bottom of Bag A.

1. With the **right gripper**, hold Bag A open at its far handle and top edge.
2. With the **left gripper**, move the first heavy item, then the second, from the staging tray into Bag A.
3. Lower each item to the bag bottom before releasing it.
4. Keep the right gripper on the bag from the first pick until the last item is supported.

Check: both heavy items sit flat on the bottom and stay still. Reposition an unstable item once before continuing.

Step 2: Load the medium layer of Bag A

Goal: the two assigned medium items rest above the heavy layer.

1. Keep Bag A open with the **right gripper**.
2. With the **left gripper**, place the first medium item, then the second, above the heavy items.

Check: neither medium item is under a heavy item, standing on edge, or pressing the bag wall outward. Reseat one failed item before continuing.

Step 3: Load the fragile layer of Bag A

Goal: the two assigned fragile items sit protected on the top layer.

1. Keep Bag A open with the **right gripper**.
2. With the **left gripper**, place the first fragile item, then the second, side by side on top of the medium layer.

Check: both fragile items are upright or flat as staged, undeformed, and free of pressure from the bag wall. Nothing rests above them and the fold band is clear.

Step 4: Fold Bag A

Goal: the bag top is folded twice without pressing on the contents.

1. Release the bag and confirm the fold band above the contents is clear.
2. With both grippers, grasp opposite top edges of Bag A.
3. Fold the flexible cloth top down about one gripper-width without pressing the contents.
4. Fold it down again by the same depth.

Check: both folds are even, no item or handle is caught in a fold, and the bag is not twisted, torn, or bulging open.

Step 5: Move Bag A to the output zone

Goal: the folded bag stands upright in its marked position.

1. With both grippers, grasp opposite bag walls below the fold.
2. Lift Bag A by its side walls, not by its handles or fold.
3. Carry it upright to the left marked output position.
4. Lower its base flat and release when stable.

Check: the base is flat, the fold stays closed, and the bag stands for 2 seconds. The center workspace is now clear.

Step 6: Build Bag B the same way

Goal: Bag B is loaded, folded, and placed exactly as Bag A was, using the remaining items.

Repeat Steps 1 to 5 on Bag B, holding it open with the **right gripper** and loading with the **left**. Bag B takes what is left on the tray, so three of its counts differ:

1. **Heavy layer (Step 1):** load the third, fourth, then fifth heavy item — **three items, not two.**
2. **Medium layer (Step 2):** load the third medium item — **one item, not two.** It alone carries both fragile items, so seat it flat across the heavy layer before releasing it.
3. **Fragile layer (Step 3):** load the third, then the fourth fragile item, side by side.
4. **Fold (Step 4) and move (Step 5)** are unchanged, except that Bag B goes to the **right** marked output position, beside Bag A.

Check: run the same checks as Steps 1 to 5. The staging tray is now empty, Bag A is left of Bag B, both bases are flat, both folds are closed, and both bags stand for 2 seconds.

Step 7: End the episode

1. Confirm all twelve items are bagged and the staging tray is empty.
2. Confirm both bags pass the Step 5 check.
3. Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

Reset with recording off.

1. Remove both bags from the output zone and unfold them.
2. Return each item to its marked row and position, in its staged orientation.
3. Replace any torn cloth bag or damaged item.
4. Restage two open bags and run Setup.

SOP violations

These actions break the SOP and are reviewed in the side-by-side review tool.

How to record a violation

For each violation, record:

- the **start timestamp**;
- the **violation name** exactly as written; and
- the **SOP rule broken**.

Use the most specific label. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag the violation and retain the episode unless project policy excludes it. Record system and material failures separately.

Violations

Violation: Required sequence not followed

- **Visible cue:** work on Bag B begins before Bag A is folded and placed in the output zone; a medium or fragile item is loaded before the layer below it is complete; a higher-numbered position is loaded before a lower-numbered one in the same bag; or a bag is folded before all its items are loaded.
- **SOP rule broken:** Steps 1–7 (finish Bag A through Step 5, repeat for Bag B in Step 6, then end; heavy, medium, fragile within each bag, lower position first).
- **Coaching note:** finish one bag completely before starting the next.

Violation: Prescribed arm, gripper, or tool assignment not followed

- **Visible cue:** the right gripper moves an item; the left gripper stabilizes a bag; or a fold or filled-bag move uses only one gripper.
- **SOP rule broken:** Steps 1 to 6, left moves items, right stabilizes, and both grippers fold and move filled bags.
- **Coaching note:** use the assigned gripper or two-gripper action.

Violation: Object handled or transported incorrectly

- **Visible cue:** an item or bag is dragged, crushed, tilted when prohibited, forced into the bag, or released unstable.
- **SOP rule broken:** Steps 1 to 6, lift, carry clear, lower gently, and release only when supported.
- **Coaching note:** keep the object controlled from pickup through placement.

Violation: Item reoriented in the air

- **Visible cue:** the gripper turns, tips, or rolls an item while it is off the tray instead of setting it down and regrasping.
- **SOP rule broken:** Steps 1 to 6, set the item flat, release, and regrasp to change orientation.
- **Coaching note:** put the item back on the tray before changing your grip.

Violation: Object dropped, collided, or released uncontrollably

- **Visible cue:** an item or bag falls, strikes another object or arm, or is released outside its destination.
- **SOP rule broken:** Steps 1–6 (maintain control and release at the assigned stable position).
- **Coaching note:** confirm the grasp and landing area before moving.

Violation: Required check, retry, or recovery not followed

- **Visible cue:** a failed check is ignored, a retry limit is exceeded, or an unwritten recovery is used.

- **SOP rule broken:** Steps 1 to 6, perform each check and use only its allowed correction.
- **Coaching note:** stop at every check before continuing.

Violation: More than one item moved at once

- **Visible cue:** two or more grocery items are lifted or carried in one pick.
- **SOP rule broken:** Steps 1–3 and 6 (move exactly one grocery item at a time).
- **Coaching note:** separate and move only the current assigned item.

Violation: Wrong item or bag selected

- **Visible cue:** an item is placed in the wrong bag, or taken from the wrong row or position — an item assigned to Bag A going into Bag B, or an item assigned to Bag B going into Bag A.
- **SOP rule broken:** Steps 1 to 3 and 6, use the fixed row position assignment for each row.
- **Coaching note:** confirm the row, position, and bag before lifting.

Violation: Heavy-to-light order not maintained

- **Visible cue:** a heavy item rests above a medium or fragile item, or a medium item rests above a fragile item.
- **SOP rule broken:** Steps 1–3 and 6 (heavy bottom, medium middle, fragile top).
- **Coaching note:** complete and check each layer before adding the next.

Violation: Active bag not stabilized during loading

- **Visible cue:** the right gripper releases the active bag while the left gripper lowers an item, or the bag collapses around the object.
- **SOP rule broken:** Steps 1–3 and 6 (hold the active bag open throughout each placement).
- **Coaching note:** keep the right hold until the object is supported.

Violation: Fragile item handled or packed incorrectly

- **Visible cue:** a fragile-row item is squeezed out of shape, dropped, placed under another item, stacked under its paired fragile item, or pressed against the bag wall.
- **SOP rule broken:** Steps 3 and 6, place fragile items gently, side by side, on top of the medium layer, with nothing above them.
- **Coaching note:** use a light grasp and keep fragile items side by side on the top layer.

Violation: Bag overfilled or fold clearance missing

- **Visible cue:** contents reach into the fold band, an item is caught in the fold area, or the fold is made despite missing clearance.
- **SOP rule broken:** Step 3 check, Step 4, and Step 6 (keep the fold band clear above the contents).
- **Coaching note:** verify the clear fold band before touching the bag top.

Violation: Bag folded incorrectly

- **Visible cue:** a fold is missing, uneven, less than full width, deeper than the clear space, or catches an item or handle.
- **SOP rule broken:** Steps 4 and 6 (fold the flexible cloth top down twice by about one gripper-width).
- **Coaching note:** align both top edges and keep the contents and handles clear.

Violation: Filled bag moved or placed incorrectly

- **Visible cue:** a bag is lifted by its handles or fold; carried tilted; placed in the wrong output position; released rocking; or left with its base not flat.
- **SOP rule broken:** Steps 5 and 6 (two-gripper side-wall grasp, upright carry, Bag A left and Bag B right).
- **Coaching note:** support both walls and lower the base flat before release.

Violation: Workspace or tool state not restored

- **Visible cue:** an item remains on the staging tray or a bag remains outside the output zone at episode end.
- **SOP rule broken:** Steps 1–7 (all items are inside the two bags and both bags are in the output zone).
- **Coaching note:** confirm the staging tray is empty and both bags are in the output zone.

Violation: Incorrect episode ending

- **Visible cue:** recording ends before both bags pass inspection, with arms away from home, with closed grippers, or without the 3-second hold.
- **SOP rule broken:** Step 7 (confirm both bags, hold, home both arms, then end).
- **Coaching note:** complete the final check before ending data collection.

Non-violation failures: log and discard an unusable episode for a recording, camera, robot, fixture, bag, or material failure; or an unexpected safety hazard.

Annotation subtasks (from SOP)

1. Load the heavy layer of a bag from the staging tray
2. Load the medium layer above the heavy layer
3. Load the fragile layer side by side on top
4. Fold the bag top down twice
5. Move the folded bag upright to its output position
6. Return both arms home and end the episode